Lecture 3

Limits

[**Note:** Remember that, the + or – signs ($x \to a^+$ or $x \to a^-$) indicate the side from which x is approaching a (indicates one sided limit). After necessary calculations, if you are able to remove the discontinuity (removable discontinuity), then you substitute exactly a in the manipulated function. If naturally without calculations or even after calculations, the discontinuity remains, then you put a greater value than a (when $x \to a^+$) or less value than a (when $x \to a^-$) to check and in order to determine whether the limit is positive infinity or negative infinity.]

C.W. $\lim_{x\to -5} \frac{(2x^2+9x-5)}{x^2-25}$ Answer: $\frac{11}{10}$

1. $$\lim_{x\to -\infty} \frac{4x^2 - x}{2x^3 - 5}$$

$$= \lim_{x\to -\infty} \frac{\frac{4x^2}{x^3} - \frac{x}{x^3}}{\frac{2x^3}{x^3} - \frac{5}{x^3}}$$ [Dividing by the highest power of x in the denominator]

$$= \lim_{x\to -\infty} \frac{\frac{4}{x} - \frac{1}{x^2}}{2 - \frac{5}{x^3}}$$

$$= \frac{0-0}{2-0}$$

$$= 0$$

H.W. $\lim_{x\to +\infty} \frac{3x+5}{6x-8}$ Answer: $\frac{1}{2}$

2. $$\lim_{x\to +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x}$$

$$= \lim_{x\to +\infty} \frac{\frac{5x^3-2x^2+1}{x}}{\frac{1-3x}{x}}$$

$$= \lim_{x\to +\infty} \frac{5x^2 - 2x + \frac{1}{x}}{\frac{1}{x} - 3}$$

$$= \frac{+\infty}{0-3}$$

$$= -\infty$$ [As $\frac{+\infty}{-c} = -\infty$]

H.W. Exercise 1.3 (13-22)

3. $\lim\limits_{x\to-\infty} \dfrac{\sqrt{5x^2-2}}{x+3}$

As $x \to -\infty$, the values of x under consideration are negative. So, we can replace $|x|$ by $-x$ where helpful.

$$\lim_{x\to-\infty} \frac{\sqrt{5x^2-2}}{x+3}$$

$$= \lim_{x\to-\infty} \frac{\frac{\sqrt{5x^2-2}}{|x|}}{\frac{x+3}{|x|}}$$

$$= \lim_{x\to-\infty} \frac{\frac{\sqrt{5x^2-2}}{\sqrt{x^2}}}{\frac{x+3}{-x}}$$

$$= \lim_{x\to-\infty} \frac{\sqrt{\frac{5x^2-2}{x^2}}}{\frac{x+3}{-x}}$$

$$= \lim_{x\to-\infty} \frac{\sqrt{5-\frac{2}{x^2}}}{-1-\frac{3}{x}}$$

$$= \frac{\sqrt{5-0}}{-1-0} = -\sqrt{5}$$

Note:

$$\sqrt{x^2} = |x| = \begin{cases} -x, x < 0 \\ +x, x \geq 0 \end{cases}$$

$$\sqrt{x^4} = \sqrt{(x^2)^2} = |x^2| = x^2$$

$$\sqrt{x^6} = |x^3| = \begin{cases} -x^3, x < 0 \\ +x^3, x \geq 0 \end{cases}$$

H.W. Exercise 1.3 (26-30)

4. $\lim\limits_{x\to+\infty} \left(\sqrt{x^6+5} - x^3\right)$

$$= \lim_{x\to+\infty} \left(\sqrt{x^6+5} - x^3\right) \cdot \left(\frac{\sqrt{x^6+5}+x^3}{\sqrt{x^6+5}+x^3}\right)$$

$$= \lim_{x\to+\infty} \frac{(\sqrt{x^6+5})^2 - (x^3)^2}{\sqrt{x^6+5}+x^3}$$

$$= \lim_{x\to+\infty} \frac{x^6+5-x^6}{\sqrt{x^6+5}+x^3}$$

$$= \lim_{x\to+\infty} \frac{5}{\sqrt{x^6+5}+x^3}$$

$$= \lim_{x\to+\infty} \frac{\frac{5}{x^3}}{\frac{\sqrt{x^6+5}}{\sqrt{(x^3)^2}} + \frac{x^3}{x^3}}$$

$$= \lim_{x\to+\infty} \frac{\frac{5}{|x^3|}}{\frac{\sqrt{x^6+5}+x^3}{|x^3|}}$$

$[\sqrt{x^6} = |x^3| = x^3 \text{ for } x > 0 \text{ as } x \to +\infty]$

$$= \lim_{x\to+\infty} \frac{\frac{5}{x^3}}{\sqrt{\frac{x^6+5}{x^6}} + \frac{x^3}{x^3}}$$

$$= \lim_{x\to+\infty} \frac{\frac{5}{x^3}}{\sqrt{1+\frac{5}{x^6}} + 1}$$

$$= \frac{0}{\sqrt{1+0}+1}$$

$$= \frac{0}{1+1} = \frac{0}{2} = 0$$

H.W. $\lim_{x\to+\infty}\left(\sqrt{x^6+5x^3} - x^3\right)$ Answer: $\frac{5}{2}$

H.W. Exercise 1.3 (31-32)

5. $\lim_{x\to+\infty}(1+2x-3x^5)$

$$= \lim_{x\to+\infty} x^5\left(\frac{1}{x^5}+\frac{2}{x^4}-3\right)$$

$$= \lim_{x\to+\infty} -3x^5$$

$$= -\infty$$

H.W. Exercise 1.3 (9-12)

6. $\lim_{x\to\infty}\sqrt[3]{\frac{2+3x-5x^2}{1+8x^2}}$

$$= \lim_{x\to\infty}\sqrt[3]{\frac{\frac{2}{x^2}+\frac{3}{x}-5}{\frac{1}{x^2}+8}}$$

$$= \sqrt[3]{\frac{0+0-5}{0+8}}$$

$$= \sqrt[3]{\frac{-5}{8}} = -\sqrt[3]{\frac{5}{8}}$$

H.W. Exercise 1.3 (24)

7. $\lim_{x\to+\infty} \frac{1-e^x}{1+e^x}$

$= \lim_{x\to+\infty} \frac{\frac{1-e^x}{e^x}}{\frac{1+e^x}{e^x}}$

$= \lim_{x\to+\infty} \frac{\frac{1}{e^x}-1}{\frac{1}{e^x}+1}$

$= \frac{0-1}{0+1}$

$= -1$

$\left[\begin{array}{c} \text{As } x \to +\infty, e^x \to 0 \\ \text{So, } \frac{1}{e^x} \to 0 \\ \text{Hence, } \lim_{x\to+\infty} \frac{1}{e^x} = 0 \end{array}\right]$

8. $\lim_{x\to-\infty} \frac{1-e^x}{1+e^x}$

$= \frac{1-0}{1+0}$

$= 1$

$\left[\begin{array}{c} [e^{-\infty} = \frac{1}{e^{\infty}} \to 0 \\ \text{So, as } x \to -\infty, e^x \to 0 \\ \text{Hence, } \lim_{x\to-\infty} e^x = 0 \end{array}\right]$

H.W. Exercise 1.3 (35-36)

9. $\lim_{x\to+\infty} \ln(\frac{2}{x^2})$

$= \lim_{x\to+\infty} (\ln(2) - \ln(x^2))$

$= \lim_{x\to+\infty} \ln(2) - \lim_{x\to+\infty} \ln(x^2)$

$= \ln(2) - (+\infty)$

$= -\infty$

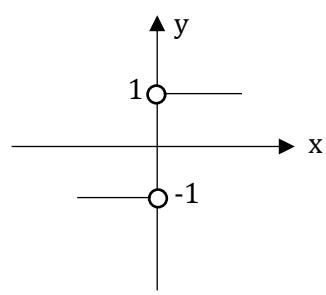
H.W. Exercise 1.3 (38)

10. Let, $f(x) = \frac{|x|}{x}$

Calculate $\lim_{x\to 0} f(x)$.

Here, $|x| = \begin{cases} +x; & x > 0 \\ -x; & x < 0 \end{cases}$

This is a piecewise function.



If x approaches 0 from the left side, the values of x are $-$ve there. When x is $-$ve,

$$f(x) = \frac{-x}{x} = -1$$

If x approaches 0 from the right side, the values of x are $+$ve there. When x is $+$ve,

$$f(x) = \frac{x}{x} = +1$$

So, $\lim_{x\to 0^-} \frac{|x|}{x} = -1$ and $\lim_{x\to 0^+} \frac{|x|}{x} = +1$.

As, L.H.L≠R.H.L, limit does not exist.

11. $\lim_{x\to 2^+} \frac{1}{|2-x|}$

As x approaches 2 from the right ($x \to 2^+$), so $x > 2$.

Here,

$$|2-x| = \begin{cases} 2-x, & 2-x > 0 \\ -(2-x), & 2-x < 0 \end{cases}$$

$$= \begin{cases} 2-x, & x < 2 \\ x-2, & x > 2 \end{cases}$$

Thus, the limit becomes,

$$\lim_{x\to 2^+} \frac{1}{x-2}$$

As $x \to 2^+$, $x-2$ approaches 0 from the positive (right) side.

Therefore, the limit is,

$$\lim_{x\to 2^+} \frac{1}{|x-2|} = +\infty$$

H.W. Exercise 1.2 (27-28)

Piecewise Function:

A function where there are multiple pieces of a function and each piece is only defined over a certain interval.

Formulas:

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \qquad \lim_{x\to 0} (1+x)^{\frac{1}{x}} = e$$

$$\lim_{x\to 0} \frac{\sin ax}{ax} = 1 \qquad \lim_{x\to 0} (1+ax)^{\frac{1}{ax}} = e$$

$$\lim_{x\to +\infty} \frac{\sin x}{x} = 0 \qquad \lim_{x\to +\infty} \left(1+\frac{1}{x}\right)^{x} = e$$

$$\lim_{x\to 0} x \sin\left(\frac{1}{x}\right) = 0 \qquad \lim_{x\to +\infty} \left(1+\frac{1}{ax}\right)^{ax} = e$$

$$\lim_{x\to 0} x \cos\left(\frac{1}{x}\right) = 0$$

12. Given:

$$f(x) = \begin{cases} 1 + 2x; & -1 \le x < 0 \\ 1 - 2x; & 0 \le x < \dfrac{1}{2} \\ -1 + 2x; & x \ge \dfrac{1}{2} \end{cases}$$

Does $\lim_{x \to 0} f(x)$ and $\lim_{x \to \frac{1}{2}} f(x)$ exist? If so, find the limits.

<u>Solution:</u>

i. Here,

$$\text{L.H.L.} = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (1 + 2x) = 1 + 0 = 1$$

$$\text{R.H.L.} = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (1 - 2x) = 1 - 0 = 1$$

As L.H.L. = R.H.L., the limit exists.
So, we have, $\lim_{x \to 0} f(x) = 1$.

ii. Here,

$$\text{L.H.L.} = \lim_{x \to \frac{1}{2}^-} f(x) = \lim_{x \to \frac{1}{2}^-} (1 - 2x) = 1 - 2 \times \frac{1}{2} = 0$$

$$\text{R.H.L.} = \lim_{x \to \frac{1}{2}^+} f(x) = \lim_{x \to \frac{1}{2}^+} (-1 + 2x) = -1 + 2 \times \frac{1}{2} = 0$$

As the one-sided limits are equal, the two-sided limit exists as $x \to \frac{1}{2}$ and it is 0.
So, we have, $\lim_{x \to \frac{1}{2}} f(x) = 0$.

13. Given:

$$f(x) = \begin{cases} (1 + 2x)^{\frac{1}{x}}; & x \ne 0 \\ e^2; & x = \dfrac{1}{2} \end{cases}$$

Find $\lim_{x \to 0} f(x)$.

<u>Solution:</u> Here,

$$\text{L.H.L.} = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (1 + 2x)^{\frac{1}{x}}$$

$$= \lim_{x \to 0^-} (1 + 2x)^{\frac{2}{2x}}$$

$$= \lim_{x \to 0^-} \left[(1 + 2x)^{\frac{1}{2x}}\right]^2$$

$$= \left[\lim_{x\to 0^-}(1+2x)^{\frac{1}{2x}}\right]^2$$

$$= e^2 \qquad \left[\text{As } \lim_{x\to 0}(1+ax)^{\frac{1}{ax}} = e\right]$$

$$\text{R.H.L.} = \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+}(1+2x)^{\frac{1}{x}}$$

$$= \lim_{x\to 0^+}(1+2x)^{\frac{2}{2x}}$$

$$= \lim_{x\to 0^+}\left[(1+2x)^{\frac{1}{2x}}\right]^2$$

$$= \left[\lim_{x\to 0^+}(1+2x)^{\frac{1}{2x}}\right]^2$$

$$= e^2 \qquad \left[\text{As } \lim_{x\to 0}(1+ax)^{\frac{1}{ax}} = e\right]$$

As L.H.L. = R.H.L., so the limit exists as $x \to 0$ and it is e^2.
Hence, we have, $\lim_{x\to 0} f(x) = e^2$.

14. Given, $f(x) = \frac{x}{|x|}$. Find $\lim_{x\to 0}\frac{x}{|x|}$.

Solution:

Here, $f(x) = \frac{x}{|x|}$

$$= \begin{cases} \frac{x}{x} = 1, x > 0 \\ \frac{x}{-x} = -1, x < 0 \end{cases}$$

R.H.L = $\lim_{x\to 0^+} f(x)$
$= \lim_{x\to 0^+} 1 = 1$

L.H.L = $\lim_{x\to 0^-} f(x)$
$= \lim_{x\to 0^-}(-1) = -1$

As L.H.L. $\neq$ R.H.L., the limit does not exist.

H.W. Exercise 1.2 (31-32)